

Faculty of Computer Science and Artificial Intelligence  
Shenzhen University of Advanced Technology

FCAB333: Computer Vision (Fall 2026)  
Homework 1

Received: Thursday September 10

Due: **8:30am, Thursday September 17, in the class**

**Note:**

1. While discussion on the course material is encouraged, you are required to work on the homework independently.
2. To get the credit, sufficient and appropriate intermediate steps are required in your answers or proofs.
3. Turn in the homework on time. Late homework will not be accepted.

1. For the following image  $A$  and filter  $B$ , find the  $5 \times 5$  filtered image  $C$  by centering  $B$  at each pixel of  $A$ . Assume pixels outside the image have intensity 0.

$A =$

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 |

$B =$

|     |     |     |
|-----|-----|-----|
| 0.1 | 0.1 | 0.1 |
| 0.1 | 0.2 | 0.1 |
| 0.1 | 0.1 | 0.1 |

2. A  $(2k + 1) \times (2k + 1)$  median filter simply replaces each pixel's value with the MEDIAN value of the pixels within a  $(2k + 1) \times (2k + 1)$  neighbourhood window centered on that pixel.
  - 1) Applying a  $3 \times 3$  median filter on the following gray-scale image, what is the resulting image? For boundary pixels, omit the pixels outside image boundary and only calculate the median value of available pixel intensities.
  - 2) What kind of noise can be reduced by using a median filter? Comparing it with the mean filter learned in the class, please describe their different application scenarios.
  - 3) Can a fixed convolution kernel implement the  $3 \times 3$  median filter for every input image? Please prove your answer.

|     |     |     |   |   |
|-----|-----|-----|---|---|
| 0.5 | 0.5 | 0.5 | 1 | 1 |
| 0.5 | 0.5 | 0.5 | 1 | 1 |
| 1   | 1   | 0.5 | 1 | 1 |
| 0.5 | 0.5 | 0.5 | 1 | 1 |
| 0.5 | 0.5 | 0.5 | 1 | 1 |

3. A 2D,  $(2k + 1) \times (2k + 1)$  kernel  $H$  is separable if it can be written as an outer-product  $H = u \cdot v^T$  where  $u$  and  $v$  are 1D (column) kernels, each with  $(2k + 1)$  elements. Please show that the mean filter, e.g.,

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

is a separable filter by finding the corresponding 1D column kernels  $u$  and  $v$ . In addition, prove that convolving an image  $I$  with a separable kernel  $H$  equals convolving each row of image  $I$  with  $v^T$  and then each column of the result with  $u$ . Use zero padding and retain the original image size at each stage. These two 1D passes significantly reduce the computational complexity.

4. Using Matlab (or other image processing software you choose) to test at least THREE edge detectors (Canny must be included) on at least one image (selected by yourself). Compare their results and describe each's advantages and disadvantages. Check the university email on August 24 for the use of Matlab license (see screenshot below). You need to print the code, input image(s) and output edge detection results, and submit all of them together with your analysis and answers, to get the credit of this homework question.

#### 信息中心

收件人 全体教职工, 全体本科生, 全体研究生

8/24 周一 下午 5:39 详情

各位老师、同学：

深圳理工大学专属MATLAB资源门户网站已上线，所有学校授权用户可通过以下网址访问深圳理工专属MATLAB在线资源（如图）

<https://ww2.mathworks.cn/academia/tah-portal/shenzhen-university-of-advanced-technology-31783433.html>



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